

MAKE UP / SUPPLEMENTARY SEMESTER EXAMINATIONS – DECEMBER 2023

Program	: B.E. – Common to all Programs	Semester	: I / II
Course Name	: Design Thinking	Max. Marks	: 100
Course Code	: AECC17 / AECC27	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- What is design thinking? Give an example of a business using Design thinking. CO1 (04)
 - Why is design thinking important in education? Explain. CO1 (04)
 - What are the six key mind sets required by designer to use design thinking to its fullest potential. CO1 (12)
- Elaborate on how does the liberatory design process ensure that designers focus on equity and inclusion in their designs. CO1 (10)
 - What is beginner's mindset? Why is it important for students to gain beginner's mindset when empathizing with users? CO1 (10)

UNIT - II

- Enumerate the five phases of design thinking. CO2 (05)
 - Explain the importance of various principles and rules to be followed in an interview. CO2 (07)
 - Prepare a questionnaire for collecting information from public to improve lakes in Bangalore. CO2 (08)
- Your college is redesigning its website. Current students are the main users of the website. Prepare a questionnaire for collecting the inputs from target users. CO2 (10)
 - Explain the importance of self-awareness and partnership in design thinking process. CO2 (05)
 - Identify one target users for the following items: CO2 (05)
 - Bicycle
 - Brush
 - Ragi
 - Drilling tool
 - Cotton.

UNIT - III

- What is convergent thinking? Why is it important and at what stage should one have it? CO3 (05)
 - What are the critical goals during prototyping? CO3 (05)
 - What five tests would you perform for testing a newly made acupuncture slippers? CO3 (10)

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6. a) Maitreyi suggested that the charges for car parking be increased substantially in the office parking lot, to encourage employees to use the office bus and other public transportation. Can you give five examples of building by 'plus'ing on the idea of Maitreyi? CO3 (05)
- b) What is divergent thinking? Why is it important? CO3 (05)
- c) Generate atleast three different engineering solutions to ensure safety in train signals and train track changer. CO3 (10)

UNIT- IV

7. a) What are the ways by which we can embrace failures? CO4 (06)
- b) What is the end goal of testing stage in design thinking? How is testing without end user conducted? CO4 (08)
- c) What is iteration? Explain with an example. CO4 (06)
8. a) Justify testing in design thinking. How is user testing conducted? CO5 (10)
- b) Case: A medical instrument manufacturing company, Aadarsh Instruments, located in Ambala, runs an export business of medical microscope. This firm is producing different types of microscopes which they export to various countries around the globe. All of the products manufactured here are conforming to the international standards. It is an ISO certified company. One of their model SL250 have a component named Focus Adjustment Knob for Slit Lamp in microscope. This microscope has found application in the field of eye inspection.
- Elaborate of applying value engineering to the focus adjustment knob.

UNIT - V

9. a) Elaborate on AMUL strategy during covid and explain it's importance with data driven design. CO5 (08)
- b) When you buy a product, we check its manufacturing or expiry date. But we rarely check the ingredients. For example, most shampoos and soaps contain sodium lauryl sulfates (SLS) which is carcinogenic, most tooth paste has very high sugar content, deodorants contain aerosols which deplete the ozone layer and harm the skin. Your task is to make aware of the customer about the importance of these ingredients. How will you give a solution to the above problem using design thinking methodology. CO5 (12)
10. a) India is chiefly an agricultural land. The cultivation of crops depends on a proper supply of water throughout the year. From olden times, large parts of our country have suffered from periods of too much rain alternating with period of drought. People have known that if the excess waters of flood seasons can be stored away for use in the fields during the dry seasons, the problem will be solved. Unfortunately, they have had neither the knowledge nor the means to do much in this matter. What little they knew have tried to put into practice. Use design thinking approach to provide some innovative solution. CO5 (08)
- b) Enumerate on Smart villages and futuristic farms. CO5 (12)
